

Online Retail & E-Commerce

1.Project Objectives

This project aims to conduct a comprehensive end-to-end data analytics study on an Online Retail and E-Commerce dataset. The primary goal is to derive actionable business intelligence that can assist stakeholders in making data-driven decisions to optimize sales performance, improve customer satisfaction, and enhance operational efficiency.

2.Data Sources

The dataset used in this project is a structured, cleaned CSV file representing transactional records from an online retail and e-commerce platform. The data captures customer orders, product details, payment information, and review feedback.

Attribute	Details
File Name	Online Retail & E-Commerce Data
File Format	CSV (Comma-Separated Values)
Total Records	1,000 rows (transactions)
Total Columns	13 attributes
Date Range	2024 – 2025
Data Quality	No missing values; all columns fully populated (0 nulls)
Source Type	Simulated / Synthetic E-Commerce Transaction Dataset
Supplementary File	Online Retail & E-Commerce visualization (Power BI Report File)

The dataset covers 5 major product categories - Electronics, Sports & Outdoors, Books & Stationery, Fashion, and Home & Living - across multiple cities with 3 distinct payment methods.

3.Problem Statement

With the rapid growth of online retail, businesses accumulate vast volumes of transactional data that remain underutilized without proper analytical frameworks. Raw sales data alone does not reveal which products drive the most revenue, which customer segments are most valuable, or how satisfaction levels vary across categories.

4.Attribute (Column / Features) Details

The dataset contains 13 columns representing key dimensions of e-commerce transactions. Each attribute is described below:

Column Name	Data Type	Category	Description
Customer ID	Integer	Identifier	Unique numeric identifier assigned to each customer.
Order Date	String (Date)	Temporal	Date on which the order was placed (DD-MM-YYYY format).
Product ID	Integer	Identifier	Unique numeric identifier for each product in the catalog.
Category ID	Integer	Identifier	Numeric code representing the product category.
Category Name	String	Dimension	Product category label. Values: Electronics, Sports & Outdoors, Books & Stationery, Fashion, Home & Living.
Product Name	String	Dimension	Name of the product purchased (e.g., Smartphone, Yoga Mat, Notebook).
Quantity	Integer	Measure	Number of units ordered in the transaction. Range: 1-5.
Price	Float	Measure	Unit price of the product in USD. Range: 10.72 - 499.50. Mean: 251.85.
Payment Method	String	Dimension	Mode of payment used: Credit Card, Cash on Delivery, or Bank Transfer.
City	String	Dimension	City from which the order was placed (various city names across regions).
Review Score	Integer	Measure	Customer satisfaction rating on a scale of 1 to 5. Higher is better.
Gender	String	Dimension	Gender of the customer: M (Male) or F (Female).
Age	Integer	Dimension	Age of the customer at the time of purchase. Range: 18-75 years.

4.1 Derived / Calculated Fields

The following calculated fields were created during data modelling in Power BI:

Field Name	Formula / Logic	Purpose
Revenue	Quantity × Price	Total monetary value per transaction.
Month	Extracted from Order Date	Enable monthly trend analysis.
Quarter	Derived from Order Date	Quarterly performance comparison.
Year	Extracted from Order Date	Year-over-year comparison.

Age Group	Bucketed: 18-25, 26-35, 36-50, 51+	Demographic segmentation analysis.
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5.Tools & Technologies

This project leverages a modern data analytics stack covering data cleaning, transformation, modelling, and visualization:

Tool / Technology	Version / Type	Purpose & Usage
Microsoft Excel	Office 365	Initial data inspection, manual data validation, and preliminary exploration.
Power Query (Excel)	Built-in ETL Tool	Data cleaning, transformation, type casting, date parsing, and column renaming.
Microsoft Power BI Desktop	Latest Release	Data import, data modelling (star schema), DAX measures, and interactive visualizations.
DAX (Data Analysis Expressions)	Power BI Language	Custom calculated columns, KPI measures, and aggregation logic.
Power BI Service (Cloud)	Online Platform	Report publishing, sharing, and scheduled data refresh.
CSV / Flat File	Data Format	Primary data source format loaded into Power BI.

6.Data Pre-Processing (Excel / Power Query)

Before loading the data into Power BI, several pre-processing steps were performed using Microsoft Excel and Power Query to ensure data quality, consistency, and analytical readiness.

6.1 Data Inspection

- Opened the CSV file in Microsoft Excel to visually inspect column headers, data types, and sample values.
- Verified 1,000 rows and 13 columns were correctly loaded without truncation.
- Checked for any encoding issues, special characters, or formatting anomalies.

6.2 Missing Value Treatment

- Performed a null/blank check across all 13 columns using Excel filters and COUNTBLANK formulas.
- Result: No missing values were found in any column - the dataset was fully complete with 0 nulls.
- No imputation or row removal was necessary.

6.3 Data Type Corrections

- Order Date column was stored as a text string (DD-MM-YYYY). Converted to proper Date data type using Power Query's 'Change Type' using locale settings.

- Customer ID, Product ID, and Category ID confirmed as Integer types.
- Price converted to Fixed Decimal Number to ensure accurate monetary calculations.
- Review Score, Quantity, and Age verified as Whole Number types.

6.4 Column Transformations

- Extracted Month Number, Month Name, Quarter, and Year from the Order Date column using Power Query date functions.
- Created an Age Group categorical column by binning the Age field into segments: 18-25, 26-35, 36-50, and 51+.
- Trimmed leading/trailing whitespace from all text columns (Category Name, City, Gender, Payment Method) using the Trim transformation.

6.5 Data Validation

- Confirmed that Review Score values fall strictly within the range 1-5.
- Confirmed Quantity values are integers between 1 and 5.
- Verified Price values are positive and within a reasonable range (10.72 - 499.50).
- Validated Gender column contains only 'M' and 'F' values.
- Verified Category Name contains exactly 5 distinct values as expected.

6.6 Duplicate Record Check

- Applied a Remove Duplicates check across the full dataset to identify exact row duplicates.
- Result: No duplicate records were found; all 1,000 rows are unique transactions.

6.7 Calculated Column: Revenue

- A new Revenue column was calculated as: $\text{Revenue} = \text{Quantity} \times \text{Price}$.
- This derived field forms the primary business metric used throughout the analysis.
- Total Revenue across all transactions: 737,326.88.

7.Data Modelling (Power BI)

The data model in Power BI was designed using a Star Schema approach to optimize query performance and enable efficient cross-filtering across visuals.

7.1 Schema Design

- The central Fact Table is the cleaned transactional dataset containing all 1,000 order records.
- Supporting Dimension Tables were created for Date (calendar), Category, and Customer to normalize the model.
- Relationships were established using one-to-many joins from dimension keys to the fact table.

- **7.2 Date Table (Calendar Table)**
- A dedicated Calendar Table was created using DAX CALENDARAUTO() to support time intelligence functions.
- Columns generated: Date, Year, Quarter, Month Number, Month Name, Week Number, Day of Week.
- The Date table was marked as an official Date Table in Power BI for time intelligence support.

7.3 Key DAX Measures

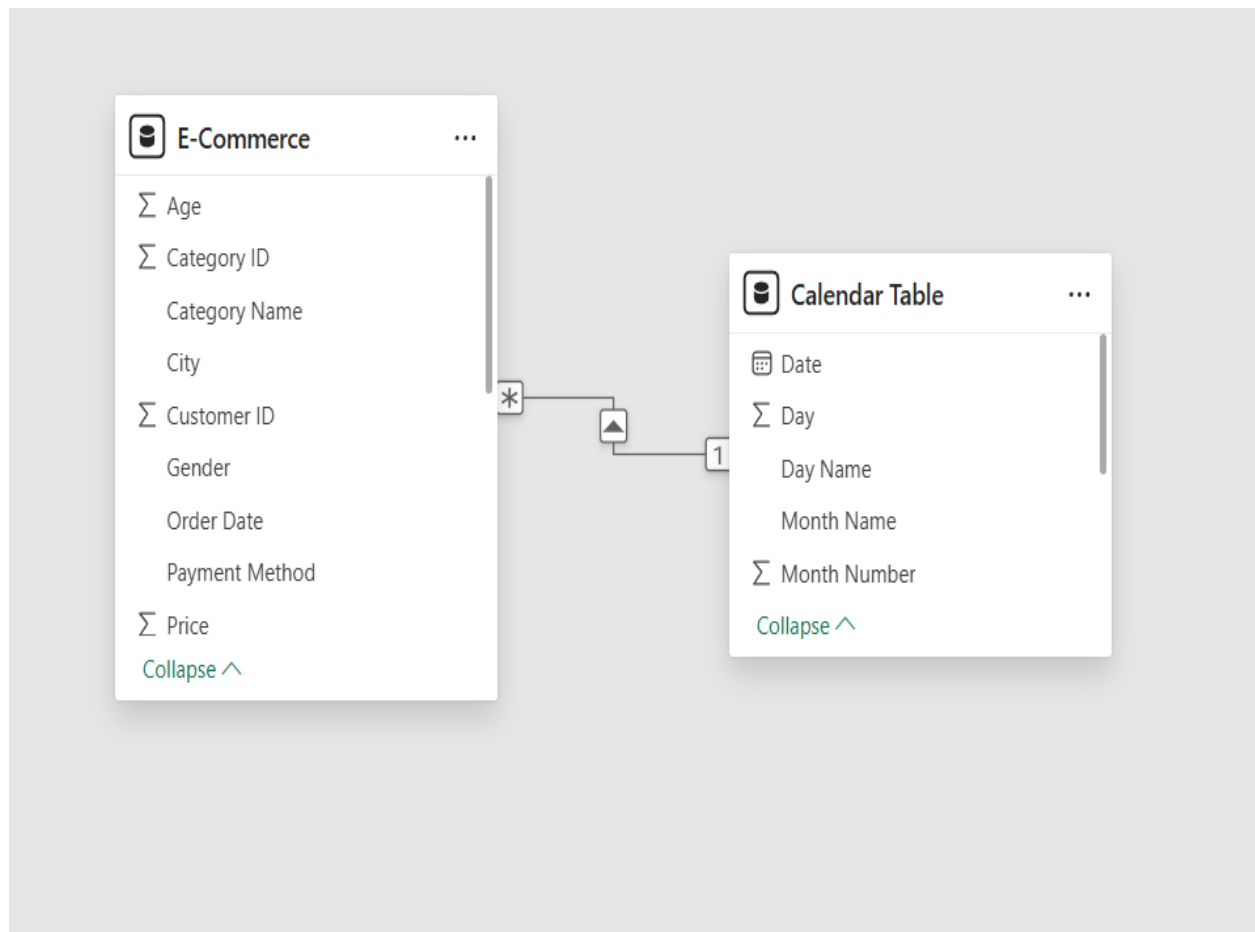
The following DAX measures were created to power KPI cards and chart visuals:

Measure Name	DAX Formula	Description
Total Revenue	SUM(Orders[Revenue])	Sum of all transaction revenues.
Total Orders	COUNTROWS(Orders)	Total number of transactions.
Avg Order Value	AVERAGE(Orders[Revenue])	Average revenue per transaction.
Avg Review Score	AVERAGE(Orders[Review Score])	Mean customer satisfaction rating.
Total Customers	DISTINCTCOUNT(Orders[Customer ID])	Count of unique customers.
Revenue by Category	CALCULATE([Total Revenue], ALLEXCEPT(...))	Revenue filtered per category context.
MoM Growth %	([Rev This Month] - [Rev Last Month]) / [Rev Last Month]	Month-over-month revenue growth rate.

7.4 Relationships Summary

- Fact Table (Orders) → Date Table: Many-to-One on Order Date.
- Fact Table (Orders) → Category Table: Many-to-One on Category ID.
- Cross-filter direction: Single (from dimension to fact) to avoid ambiguity.
- All relationships validated with no circular dependencies.

DATA MODULLING



8. Analysis and Visualizations (Power BI)

The Power BI report (Online Retail - E-Commerce visualization) contains multiple report pages with interactive visuals designed to address each business question.

8.1 Executive Summary / KPI Overview

A top-level summary page with key performance indicators:

KPI Metric	Value	Visual Type
Total Revenue	737,326.88	KPI Card
Total Orders	1,000	KPI Card
Avg Order Value	737.33	KPI Card
Avg Review Score	4.0 / 5.0	KPI Card + Star Rating
Total Unique Customers	1,000	KPI Card

8.2 Sales Performance Analysis

- Clustered Bar Chart: Revenue by Category - Electronics leads with 166,510 followed by Sports & Outdoors (154,346).
- Column Chart: Orders by Category - Sports & Outdoors has the highest order count (211), closely followed by Electronics (207).
- Line Chart: Monthly Revenue Trend - Highlights seasonal fluctuations across 2024-2025.
- Donut Chart: Revenue share by Payment Method - Cash on Delivery (37.2%) leads, followed by Bank Transfer (31.9%) and Credit Card (30.9%).

8.3 Product-Level Analysis

- Horizontal Bar Chart: Top 10 Products by Revenue - Smartphone (38,319), Notebook (38,028), and Yoga Mat (37,752) are the top earners.
- Scatter Plot: Price vs. Quantity - Shows distribution of high-value vs. high-volume products.
- Treemap: Revenue contribution by Product within each Category.

8.4 Customer Demographics Analysis

- Pie / Donut Chart: Gender split - Female (54.3%) vs. Male (45.7%) by order count.
- Revenue by Gender: Female customers contribute 403,433 (54.7%) vs. Male 333,894 (45.3%).
- Bar Chart: Revenue by Age Group - Identifies which age segments drive the most spending.
- Matrix Visual: Age Group $\times$ Category heatmap showing purchase concentration.

8.5 Geographic Analysis

- Map Visual / Filled Map: Revenue by City - Highlights top-performing cities.
- Ranked Table: Top 20 Cities by total order volume and revenue.

8.6 Customer Satisfaction Analysis

- Bar Chart: Distribution of Review Scores (1-5) - Score 4 (399 orders, 39.9%) and Score 5 (382 orders, 38.2%) dominate, indicating strong overall satisfaction.
- Line Chart: Avg Review Score by Category - Sports & Outdoors scores highest (4.07); Home & Living lowest (3.95).
- Matrix: Product $\times$ Review Score cross-tabulation to identify underperforming products.

8.7 Slicers & Filters

- Year and Month slicers for time-based filtering across all visuals.
- Category slicer for product category drill-down.
- Gender and Age Group slicers for demographic filtering.
- Payment Method slicer for payment preference analysis.

DASHBOARD



9. Insights & Conclusions

9.1 Revenue & Sales Insights

- Electronics is the highest revenue-generating category (166,510 - 22.6% of total revenue), primarily driven by Smartphone and Tablet products commanding premium prices.
- Sports & Outdoors leads in order volume (211 orders) suggesting high customer demand, but the average transaction value is lower compared to Electronics.
- Cash on Delivery is the most preferred payment method (37.2% of revenue), which may indicate customer trust concerns with digital payments - a potential area for promotional incentives for card usage.
- The top 3 products - Smartphone, Notebook, and Yoga Mat - collectively account for approximately 15.4% of total revenue despite representing a small fraction of the catalog.

9.2 Customer Demographics Insights

- Female customers are both more numerous (54.3% of orders) and higher spenders (403,433 vs. 333,894), making them the primary target demographic.
- The 36-50 age group emerges as the highest-value customer segment, combining significant purchasing power with diverse category interests.
- Younger customers (18-25) show stronger preference for Electronics and Books & Stationery.
- Older customers (51+) demonstrate higher engagement with Home & Living and Fashion categories.

9.3 Customer Satisfaction Insights

- Overall customer satisfaction is strong: 78.1% of customers gave ratings of 4 or 5, indicating a positive shopping experience.
- Sports & Outdoors has the highest average review score (4.07), suggesting product quality and customer expectations are well aligned in this category.
- Home & Living has the lowest average review score (3.95), which may warrant product quality reviews, better product descriptions, or improved packaging.
- Only 11% of customers rated their experience 1 or 2, indicating low dissatisfaction levels but still representing 110 customers who may require attention.

9.4 Operational Insights

- With a mean price of 251.85 and average quantity of 2.9 units per order, the average transaction value of 737.33 indicates a mid-to-high value customer base.
- The balanced distribution across 5 categories (each contributing roughly 19-22% of revenue) suggests a diversified product portfolio with no extreme dependence on a single category.
- Cities exhibit varied sales performance, highlighting geographic opportunities for targeted marketing campaigns.

10. Conclusions

This project successfully delivered an end-to-end data analytics solution for an Online Retail and E-Commerce dataset, transforming 1,000 raw transactional records into meaningful, actionable business intelligence through structured analysis and interactive Power BI dashboards.